Solution Requirements

Date	27 june 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

This document defines the functional and non-functional requirements of the HDI Prediction System. Functional requirements describe the features and operations performed by the system, while non-functional requirements define quality attributes such as performance, security, reliability, and usability. These requirements ensure the system delivers accurate HDI predictions and a smooth user experience.

Functional Requirements (FR): Functional requirements describe the features and operations that the HDI Prediction System must perform, such as accepting socio-economic data, processing it using a machine learning model, and predicting the Human Development Index (HDI).

Non-Functional Requirements (NFR): Non-functional requirements describe the quality standards the HDI Prediction System should maintain, including performance, reliability, usability, security, and maintainability, to ensure accurate predictions and efficient system operation. **Step 1: Functional Requirements (FR)**

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in using username and password.	High

2	Authorization levels	Admin manages data; users can view and predict HDI.	High
3	External interfaces	Connects with datasets and the machine learning model.	High
4	Transactions processing	Accepts input, processes data, and predicts HDI.	High
5	Reporting	Displays HDI prediction results and downloadable reports.	Medium
6	Business rules, etc.	Validates all required inputs before prediction.	High
7	Compliance to laws or regulations	Protects user data and follows privacy standards.	Medium

8	*Other*	Allows users to compare previous and current predictions.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate predictions quickly.	Response time < 3 seconds
2	Scalability	Support multiple users simultaneously.	Supports 500+ users
3	Security & Data Privacy	Protect user data using secure storage.	Encrypted data and secure login
4	Reliability & Availability	System should be available whenever needed.	99% uptime
5	Usability & Accessibility	Simple and user-friendly interface.	Easy navigation and responsive UI

6	*Other*	Easy maintenance and future model updates.	Modular and maintainable code